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Peace-Work-Fatherland

MINISTRY OF HIGHER
EDUCATION



INTERNSHIP REPORT

MANUFACTURING OF A FLANGE FOR A CONCRETE MIXING MACHINE

A report of internship carried out at NOUGNING SARL from 28th of June to 31st of July for the partial fulfillment of requirement for the award of the Bachelor of Technology (B-TECH)

Speciality :

MECHANICAL MANUFACTURING

Presented by :

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ACADEMIC YEAR 2023-2024

APPROVAL

I, the undersigned **MR. BESONG PLATINI** do hereby acknowledge and testify that this internship report entitled **MANUFACTURING OF A FLANGE, CASE OF A CONCRETE MIXING MACHINE** is the original work of TCHEMO ISRAEL JUNIOR written after an elaborate research in the field of mechanical manufacturing.

The responsibility for any errors, mistakes and omissions, and facts that may be found in this work is that of the researcher. My responsibility has been to ensure that the report conforms to international standard and format required by the FOMIC POLYTECHNIC UNIVERSITY under the supervision of the Ministry of Higher Education.

I find this report convincing and as such fits for public defense before the competent jury for the evaluation in partial fulfilment of the requirement for the award of Bachelor of technology (B-TECH) in Mechanical Manufacturing.

ACADEMIC SUPERVISOR

MR. BESONG PLATINI

Signature.....

Date.....

CETIFICATION

The internship report entitled the manufacturing of a flange done by TCHEMO ISRAEL JUNIOR, meets the requirement and regulation governing the award of the Bachelor of Technology (B-TECH) in Mechanical Manufacturing by FOMIC POLYTECHNIC under the tutelage of the Ministry of Higher Education, hence approve for its contribution to scientific knowledge and literacy presentation.

ACADEMIC SUPERVISOR

PROFESSIONAL SUPERVISOR

MR. BESONG PLATINI

MR. TCHUTZO NGUMETE BORIS

Signature.....

Signature.....

Date.....

Date.....

PRESIDENT OF JURY

Signature.....

Date.....

DEDICATION

TO

MY LOVELY FAMILY

ACKNOWLEDGMENTS

This report is the result of a teamwork and extensive collaboration. We cannot present it without showing our gratitude to all people including my father, my sources and all those who directly or indirectly contributed in a decisive way to the success of this work.

I'm great full to:

- **Mr. BESONG PLATINI** my academic supervisor for all his support and assistance for the success.
- **Mr. TCHUTZO NGUMETTE BORIS** my professional supervisor for his humble and great assistance offered to me.
- **Mr. FOSSUNG ROMANUS** for giving us all the necessary requirement needed to come out with this document.
- My **father Mr. TCHEMO ELIE** for his endless financial support.
- All my friends and classmates for their moral support.

ABSTRACT

This document is the output of a completion of an academic internship (which had been carried from 28th of June to 31st of July 2022) in order to blend theory with practical knowledge done at NOUGNING SARL located in LOGBESSOU DOUALA. The report is made up of the presentation of the company, the activities carried out during the internship period, difficulties encountered, suggested solutions and recommendations.

During our internship, we fabricated a flange for a concrete mixing machine. Since the one we had was bad because it was assembled with welding with poor interval of tolerance. We carried out findings and we saw that we had to fabricate another flange which will perform the same functions as the previous.

For the realization of this part, we proceeded in a methodological method which implies; making the choice of the most suitable material, coming out with the detail and manufacturing drawing of the piece, then it's manufacturing analysis. Facing, drilling, Mortising and a series of other operations were carried out on the piece. Some of the machines used may include lathe and milling machines.

At the end, when the flange and other parts where mounted, the machine functioned properly. We can say it was a great success since the production rate increased.

RESUME

Ce document est le résultat d'un stage académique (qui avait été effectué du 28 juin au 31 juillet 2022) afin de mélanger la théorie avec les connaissances pratiques faites au NOUGNING SARL situé à LOGBESSOU DOUALA. Le rapport est composé de la présentation de l'entreprise, des activités réalisées pendant la période de stage, des difficultés rencontrées, des solutions suggérées et des recommandations.

Au cours de notre stage, nous avons fabriqué une bride pour une machine à béton. Depuis celui que nous avions était mauvais parce qu'il a été assemblé par soudage avec une mauvaise intervalle de tolérance. Nous avons effectué des constats et nous avons vu que nous devons fabriquer une autre bride qui remplira les mêmes fonctions que la précédente.

Pour la réalisation de cette pièce, nous avons procédé dans une méthode méthodologique qui implique; faire le choix du matériau le plus approprié, sortir avec le détail et le dessin de fabrication de la pièce, puis, l'analyse de fabrication de la pièce. Le dressage, le forage, le mortaisage et une série d'autres opérations ont été effectués sur la pièce. Certaines des machines utilisées peuvent inclure des machines de tour et de fraisage.

À la fin, lorsque la bride et d'autres pièces sont montées, la machine fonctionne correctement. Nous pouvons dire que ça été un grand succès puisque le taux de production a augmenté.

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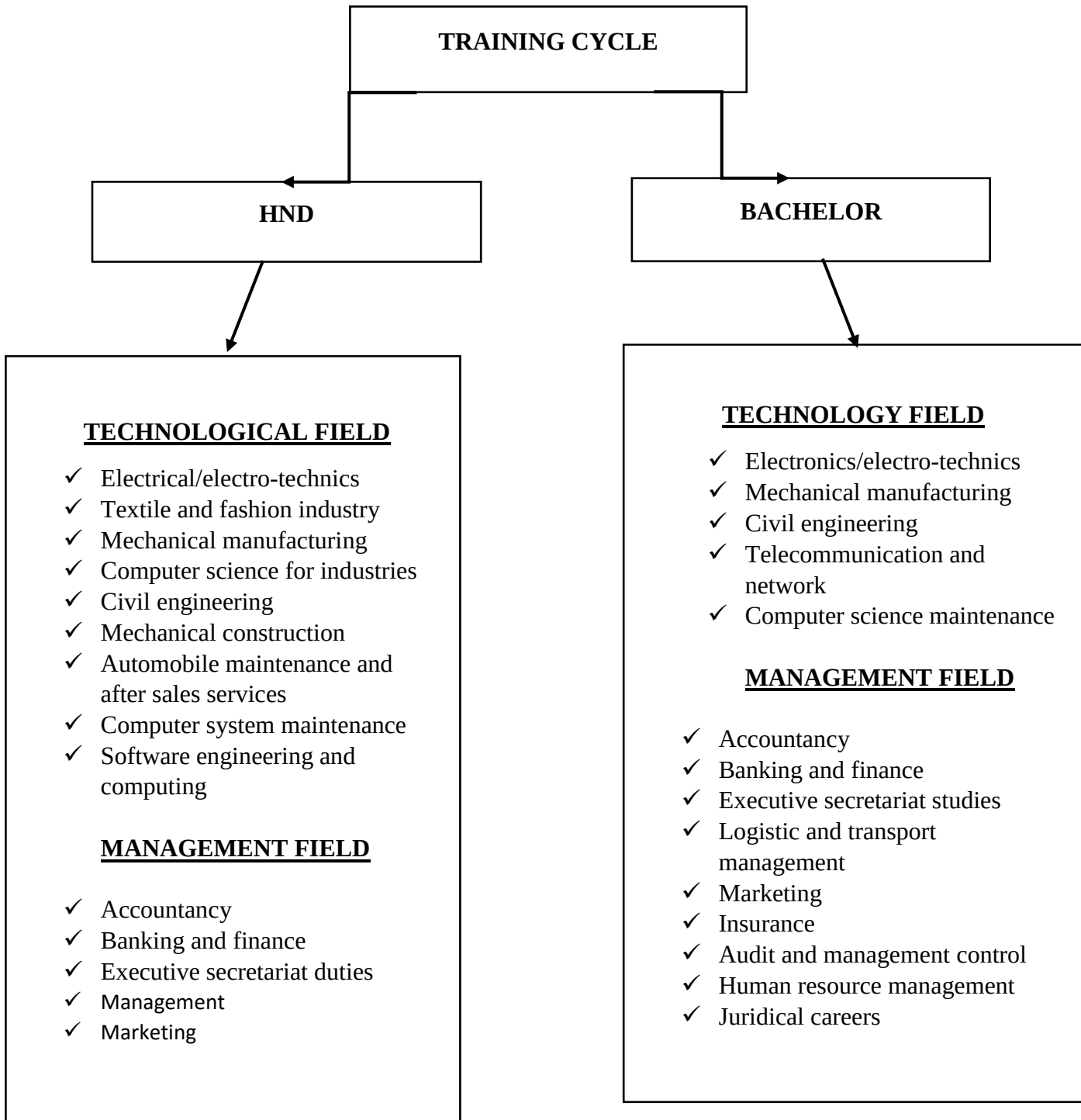
- B: rough surface.
- CS: cutting speed.
- D: cylindrical surface.
- F: flat surface.
- C: conical surfaces.
- B-TECH: BACHELOR OF TECHNOLOGY
- FOMIC POLYTECHNIC
- m_{d1} : machine dimension
- t_{d1} : tool dimension

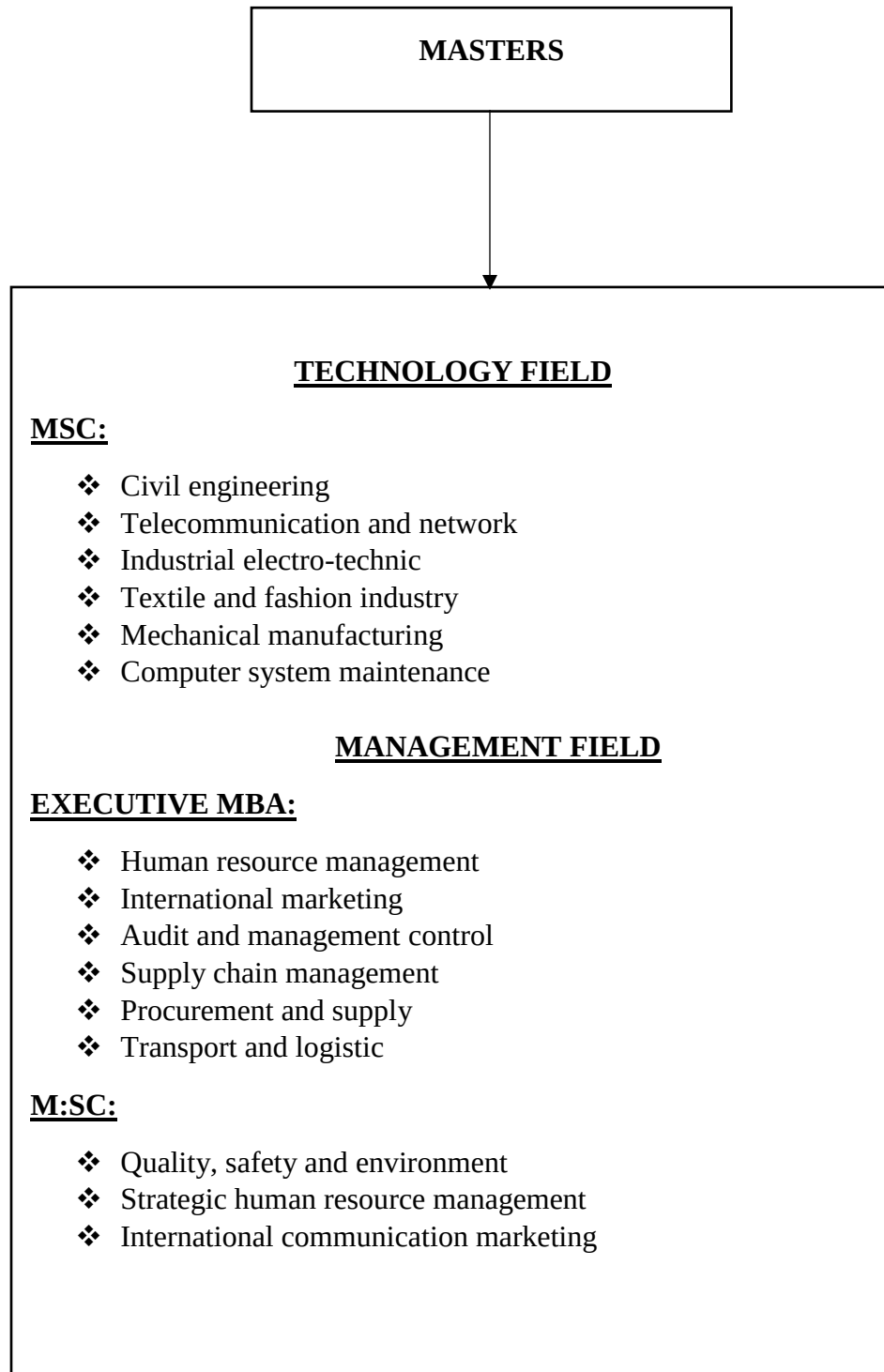
PREFACE

FOMIC POLYTECHNIC UNIVERSITY is a multidisciplinary **Institute Of Technology, Economics Science and Management** .Our strengths comes from quality of our lecturers , well equipped workshops and efforts to meet international standards . A varied offer of training in close link with research has enabled our recognition with the quality of our teaching and excellent results as far as exams and professional insertion are concern.

A TRAINING OFFER FOR EXCELLENCE

FOMIC POLYTECHNIC offers more than 30 certificates **In Technology, Economic Science and Management**. Our objective is to train students in the **HND cycle, Bachelor and MASTER Degree's** while providing them opportunities for training throughout life. For 10 years today, this has been **FOMIC** main goal. We offer strategic support to job applicant, wage-earning negociatiation in enterprises amisdst varying economic threshold.





A TECHNOLOGICAL POLE

THE GROUP FOMIC POLYTECHNIC has a technologic pole called **FOMIC TECHNOPOLE**

FOMIC TECHNOPOLE is a pole of technology with an International scope.

VALUES

FOMIC POLYTECHNIC is deeply attached to universal value creation and diffusion of knowledge, solidarity, open to the world and ideas, participation in local life, preparation of students for professional insertion and citizenship.

OUR PARTNERS

We are actively investing in knowledge's (scientific park, pole of competitiveness and excellence, as well as institutions favoring the value of research and the creation of innovation enterprises...)

To perfectly carry on our mission, FOMIC POLYTECHNIC benefits from permanent support of territorial collectivity in its developed project both nationally and internationally. It has a large network of university partners, economic, social and cultural entities with

- The University of Bamenda;
- The University Douala;
- The University of Buea;
- The University of Orléans
- The University of Evry;
- The University of Divanovo -Russia

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GENERAL INTRODUCTION

A- BACKGROUND STUDY

An academic internship is a period of our studies when students of the higher institutions are sent to various organisations, industries, companies or enterprises and associations of various field from which they blend theory to practical for a better understanding of the field. The work in this present document is the fruit of what the intern did during his internship period. we shall see some machining operations like; drilling, facing, turning. Explanations will be given on the detail and manufacturing drawing.

- Chapter 1: Presentation of the company.
- Chapter 2: Internship activities.
- Chapter 3: Situation analysis
- Chapter 4: Summary and recommendations

B- CHOICE OF CASE STUDY

This part of the report is concerned with what we did during our internship period. We did our internship at NOUGNING SARL and during this period, we worked in the manufacturing department and had to replace a flange that was already bad, brought to us by a company producing blocks. After critical analysis on the piece we decided to manufacture a new one, we changed it's material from cast iron into medium carbon steel considering into consideration the fact it had to resist to corrosion. The purpose of this flange is to grab to cylindrical shafts during rotation.

C-OBJECTIVES OF THE STUDY

At the end of the internship the intern should be able to;

- Understand the setup or organisation of a workshop
- Analysis of machined surfaces
- Dimension pieces to be manufactured or machined

CHAPTER I. PRESENTATION OF THE COMPANY

1.1 HISTORY OF THE COMPANY.

NOUGNING SARL is a private limited company situated in Douala, Littoral Region precisely at CARREFOUR BASSONG LOGBESSOU. It accept Students for Internship, training them and enabling them to have more practical skills, thereby increasing their skills in the field of Mechanical fabrication. By so doing they help the government to train young Cameroonians more operational, for either self employment etc.

NOUGNING SARL was created in the year 2008. Its deals with the fabrication and realization of mechanical parts and also machinery, almost all their material came from Cameroon. In order to reduce expenses and maximize profit, he opted in local fabrication of metallurgical products.

1.2 THE OBJECTIVES OF NOUGNING SARL

Like any other company, the major purpose or goal of NOUGNING SARL is to maximize it's profit while satisfying the needs of its customers.

1.3 PRESENTATION OF THE SERVICE

Working hours at NOUGNING SARL have been classified in the following table;

DAYS	MORNING	BREAK	AFTERNOON
MONDAY	08H00-12H00	12H00-13H00	13H00-17H00
TUESDAY	08H00-12H00	12H00-13H00	13H00-17H00
WEDNESDAY	08H00-12H00	12H00-13H00	13H00-17H00
THURSDAY	08H00-12H00	12H00-13H00	13H00-17H00
FRIDAY	08H00-12H00	12H00-13H00	13H00-17H00
SATURDAY	08H00-12H00	12H00-13H00	13H00-17H00

Table 1: TIME TABLE

1.4 IDENTIFICATION SHEET OF NOUGNING SARL

Table 2:Identification sheet of NOUGNING SARL	Carrefour bassong- Logbessou
Telephone	691-571-657/ 674-450-066
Creation date	2016
Control	M0916125857398Z
Matricule	RC/DLN/2016/B/1872/N
Social director	Douala-bassong
Number of employees	21 (Approximately)
Principal activity	INDUSTRIAL
Capital	10.000.000 FCFA (approximately)
Judicial status	S ARL

Table 1: IDENTIFICATION SHEET

1.5: ORGANISATIONAL STRUCTURE OF NOUGNING SARL

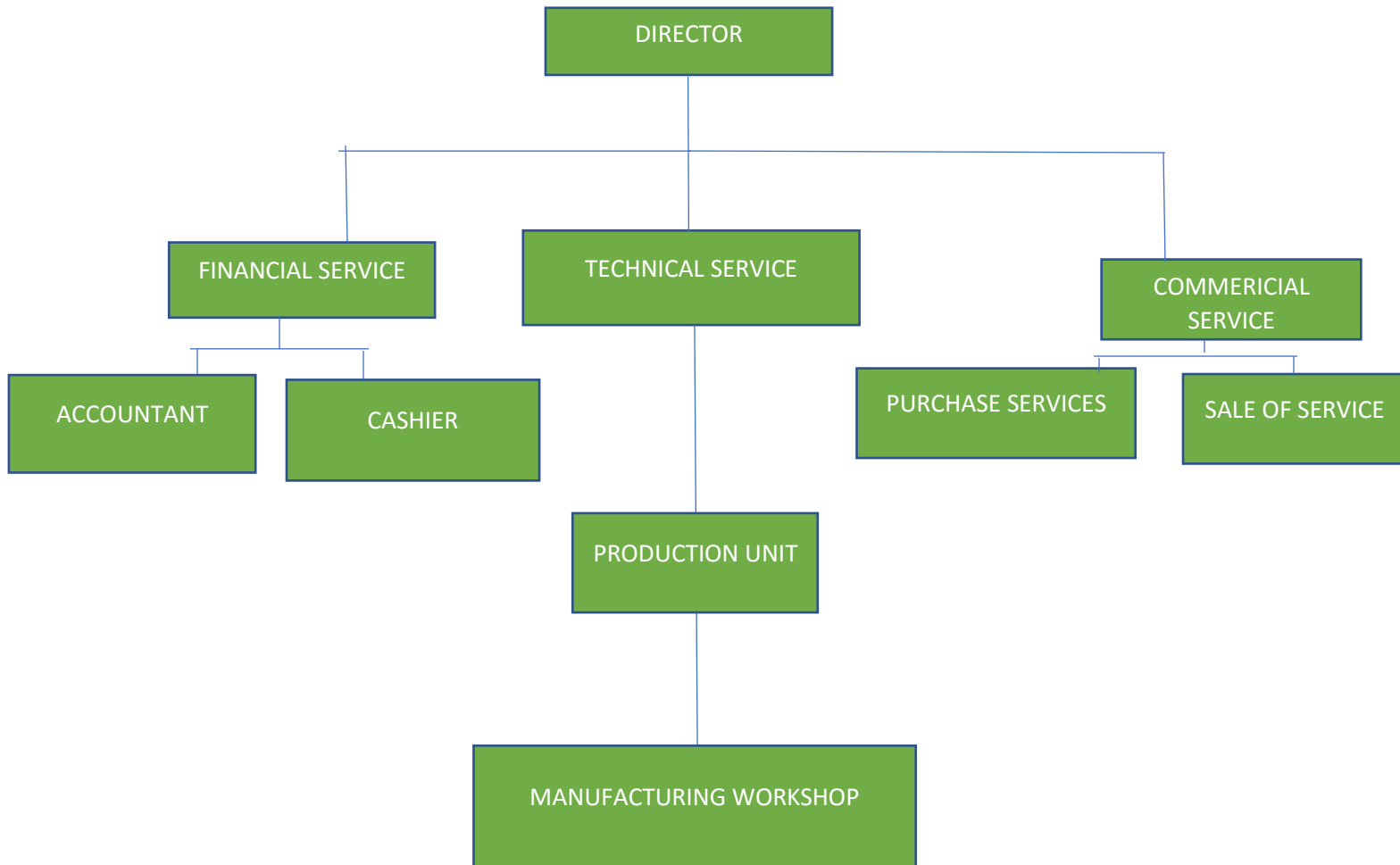
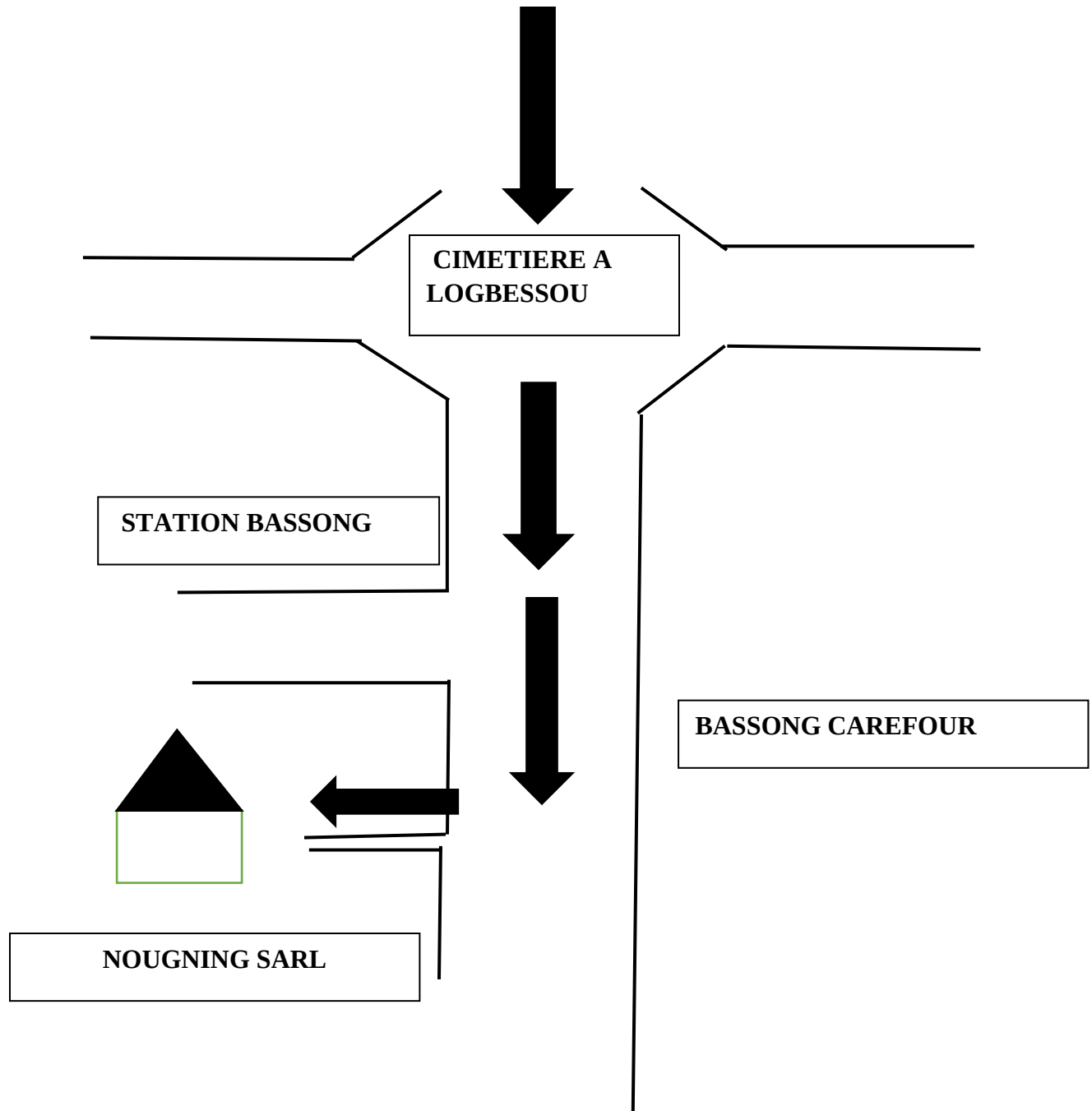


Figure 1: ORGANISATIONAL STRUCTURE

1.6 GEOGRAPHICAL LOCATION OF NOUGNING SARL



CHAPTER II. INTERNSHIP ACTIVITIES

INTRODUCTION:

Internship carried out at NOUGNING SARL was profitable. This is because most the theory and practice was well assimilated and put into practiced.

This chapter gives you a view of the activities that the intern carried out during his internship period. It range from safety precaution, organisation enterprise of machine with goals and objectives stated in chapter one.

The intern will briefly descript in this chapter the different cutting tools he used, as well as the activities.

2.1: ACTIVITIES CARRIED OUT DURING THE INTERNSHIP

During this period of internship, we carried out numerous activities in the various units of the enterprise especially in the sheet metal and fabrication workshop. These activities are enumerated in the tabulated form as per week below.

2.1.1 ACTIVITIES CARRIED OUT

WEEKS	DESCRIPTION OF WORK DONE
Week 1 From the 28th June to the 3rd of July 2022	• Presentation of the workshop. • Presentation of machines. • Meeting on safety rules in the workshop. • We were taken to the welding department so that we could ameliorate on our welding skills and to teach those who don't know how to weld or operate on a welding post. It was not a problem for me because I had solid background in sheet metal and welding in secondary school, as it was my trade. • Cleaning of the workshop.
Week 2 From the 5th July to the 10th July	• Revision on safety that is (PPE); Personal protective equipments and it's importance. • Stated working on productions of windows in the sheet metal department. • Cleaning of the workshop.

Week 3 From the 12th July to the 17th July	- Continued the production of windows. - A concrete mixer was brought to us and had a transmission problem, after analysing it we discovered flange which was giving motion to the shaft was bad. We saw that we could manufacture a coupling to replace the flange. - Cleaning of the workshop.
Week 4 From the 19th July to the 24th July	- Production of windows. - Cleaning of the workshop.
Week 5 From the 26th July to the 31st of July	- Learning of mechanical operations such as; - Drilling - Turning - Facing - Mounting of the completed windows produced. - General cleaning

Table 3: WEEKLY ACTIVITIES

2.2: LEARNING OUTCOME (BENEFITS DERIVED FROM THE INTERNSHIP)

The intern has benefited a lot from the internship which includes the following.

- We improved our skills on how to use the lathe machine.
- We improved our skills on how to read and understand the vernier calliper.
- Collaboration (team work).
- Time management.
- Understanding of some safety rules and symbols.
- Perfect use of a grinding machine.
- Discovery of some industrial precautions and safety.
- Improvement on dressing code in the workshop.
- Got to know several machines I never used before.

2.3: CHALLENGES FACED DURING THE INTERNSHIP

During my internship at NOUGNING SARL, we encountered some difficulties. Some of these include:

- **I was not fortunate to work much on lathe machine:** This was because the work in the sheet metal workshop took me so much of my time during my internship.
- **Shortage of electricity:** This has been really very difficult because there was shortage in time and delayed the working hours.

2.4: PROBLEM STATEMENT

This section of the report deals with actually made or pushed the intern to write on this particular topic. The intern carried out his internship at NOUGNING SARL in the department of mechanical manufacturing. During this period of internship, the intern carried out multiple research on the properties of various materials and discovered the suitable material for this flange was cast iron. It has the following properties; less expensive, low thermal expansion and has a high heat resistance etc. The problem to be solve here is to avoid the flange to get bad fast because the previous one was mounted by welding and was in cast iron and had much weight which had an impact on the mechanism therefore increasing the rate of wearnees and makes it very very difficult to dismountle. We decided to manufacturing a new one with material c35 (medium carbon steel) which has a function to firmly hold the shaft of the concrete mixer and transmit power to it. By doing so the manufactured flange will have the following advantage; cheaper in cost, construction will be more reliable and will be weightless.

CHAPTER III. METHODOLOGY AND ANALYSIS OF DATA

INTRODUCTION: This chapter focuses on the fabrication analysis and many machining process of a flange that helps transmit motion from one driving shaft to another driven one in a concrete mixer.

3.1: DESCRIPTION OF A LINKAGE GRAPH

1. Linkage graph: This is a graph that defines all the dimensional and geometrical links between the rough surfaces and machining surfaces or between several machined surfaces.

2. Table of machining operations: It shows the realisation of an elementary surface, linked to other surfaces by dimensional and geometrical tolerances, impose a certain number of operations which is in function of;

- The interval of tolerance of this surface
- The roughness of this surface
- The quality (Q)

3. Table of grouping of surfaces: During machining, some surfaces are obtained simultaneously either by the same tool, by the association of tools or by the use of several spindles which can receive several tools working simultaneously, this gives the possibility of grouping these surfaces and giving them a group name for the rest of the analysis.

4. Table of machining restriction: This table is established from dimensioning graph the detail drawing and the definition table of elementary operations. It shows all the geometric, technological and economic constraint of a surface.

5. Table of machining: This table shows which surface is manufactured in logical order, it consist of 2 parts; a constraint / restriction matrix and a matrix level.

6. Table of grouping phase: This a table which group all phases and their sub phases together place vertically 0, 1, 2, 3 and so on.

7. Draft of machining sequence: Gives the order of phases and the chronological order of obtaining the elementary operations in each machine phase.

8. Phase contract: The phase contract is a document established by the method office to verify the operational process of the phase considered (different stages of a machining of the part).

3.2) Choice of material

The material used to fabricate the linking system in mechanical manufacturing varies according to the type of system and the function of that system.

3.2 CRITICAL ANALYSIS OF THE IDENTIFIED PROBLEM

3.2.1 DESCRIPTION OF THE LINKING SYSTEM

The linking system is composed of a flange having a key for the attachment and guiding of parts and the external part of the flange its self which have bored holes on it for attachment. The function of the system is to maintain and guiding rigidly the mixer's part during functioning.

3.2.2 CHOICE OF THE MATERIAL

The material used to fabricate the linking system in mechanical manufacturing varies according to the type of system and the functioning of that system. The material of the flange is cast iron. We used mild steel of (C35) due to its high tensile strength, good malleability, good ductility.

3.2.3 DATA COLLECTION AND PROCESSING

Here the intern focus on calculation involved in configuring the machine used in the realization of the fabrication of the flange. The speed for turning operation, this is essential to calculate in order to avoid rapid tool wear and thus obtaining a perfect machined surface. Normally the speed of the machining for most materials have been stated by the manufacturers of the materials in the ISO standard.

Spindle speed foe turning operation

where Cs= Cutting speed

$$N = \frac{CS \times 1000}{\pi D}$$

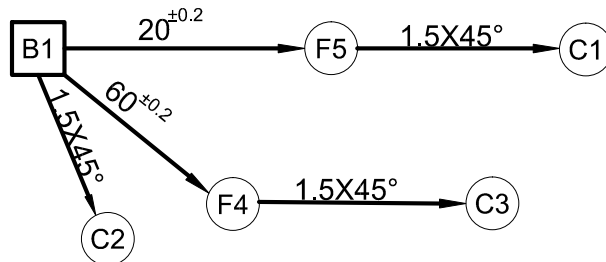
D =diameter of piece (180)

$$Cs = 27\text{m/min}$$

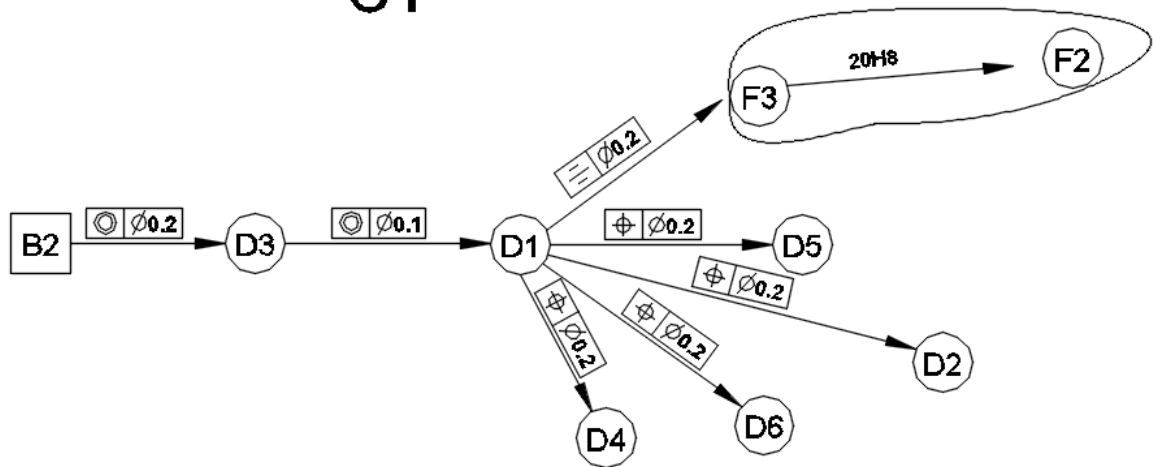
$$N = \frac{27 \times 1000}{180 \times \pi} = 47.77$$

DIMENSIONAL LINKAGE GRAPH

OX



OY



OZ

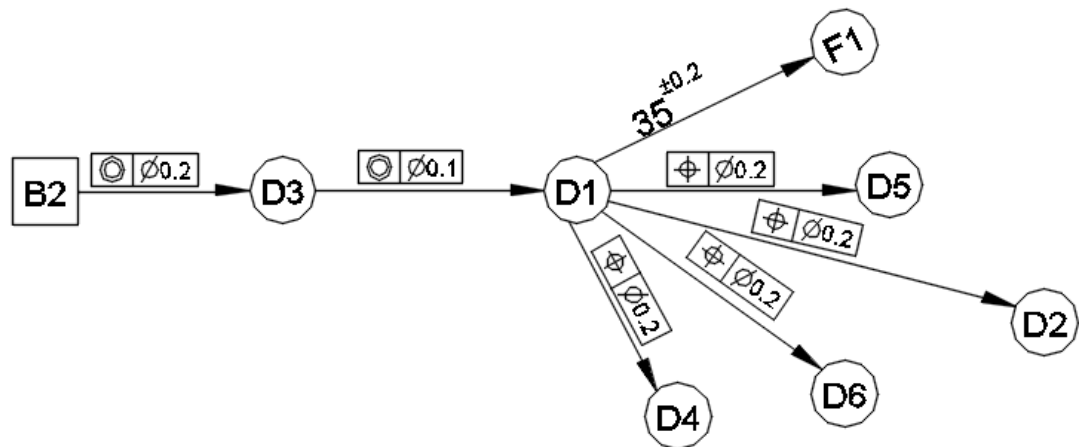


TABLE OF ELEMENTARY OPERATIONS

[illegible]

Table 4: ELEMENTARY OPTIONS

TABLE OF ASSOCIATION OF SURFACES

SURFACES TO BE ASSOCIATED	JUSTIFICATION			NEW DESIGNATION
	Cutting tool	Tool holder	Others	
F1,F2, F3	MORTISE TOOL	TOOL POST		G1
D3,F5	KNIFE TOOL	TOOL POST		G2

Table 5: ASSOCIATION OF SURFACES

TABLE OF MACHINE RESTRICTIONS

[illegible]

Figure 2: MACHINE RESTRICTION

TABLE OF MACHINE LEVELS

[illegible]

Figure 3: MACHINING LEVELS

TABLE FOR GROUPING OF PHASE

LEVEL	ELEMENTARY OPERATIONS									
1	G2F	F4F	C2F							
2	D1F		C3F			D2F, D4F, D5F,				
3			C1F	G1F		D6F				
4										
5										
6										
7										
8										
9										
10										
11										
12										
13										
14										
15										

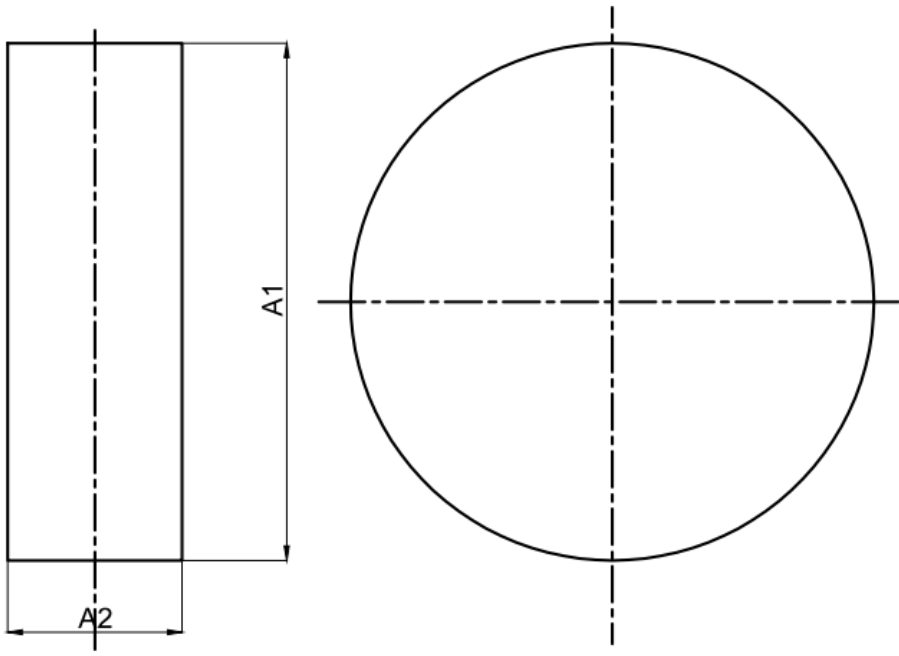
Figure 4: GROUPING OF PHASES

DRAFT OF MACHINE SEQUENCE

PHASE REF	DESIGNATION	SURFACES MACHINED
10	CONTROL OF RAW MATERIAL	
20	TURNING	F4F, G2F, C2F,C3F C1F,
30	MORTISING	G1F
40	DRILLING	D2F, D4F, D5F, D6F
50	FINAL CONTROL	
60		
70		
80		
90		
100		

Figure 5: DRAFTING SEQUENCE

MINESUP		HND Mechanical Fabrication		Part: Material: C30		
		PHASE CONTRACT				
Session: 2022						
Phace NO: 10		Machine tool used: CONTROL POST				
Referential:						
MACHINING OPERATIONS		TOOLS		CUTTING ELEMENTS		
Ref.	Description	Cutting	Verification	Cs m/mm	N Rev/min	F mm/rev
a)	General control A1=180 , A2=38		Vernier calliper 1/50			



The drawing consists of two views of a flange. The left view is a side elevation showing a rectangular profile with a vertical centerline. Dimension A1 indicates the total height, and dimension A2 indicates the width of the base. The right view is a top-down circular view of the flange, also featuring a vertical centerline.

Figure 6: PHASE CONTRACT

MINESUP		HND Mechanical Fabrication		Part:		
Session: 2022		PHASE CONTRACT		Material: C30		
Phase NO: 20		Machine tool used: KNIFE TOOL				
Referential:						
MACHINING OPERATIONS		TOOLS		CUTTING ELEMENTS		
Ref.	Description	Cutting	Verification	Cs m/mm	N Rev/min	F mm/rev
a)	Face F4F : md4=40	Knife tool	Vernier calliper 1/50	27	47.77	0.1
b)	Turning G2F: md5=90	Knife tool	Vernier calliper 1/50	27	47.77	0.1
c)	Chamfering C2F,C1F,C3F	Knife tool	Vernier calliper 1/50	27	47.77	0.1
	td3=60					

The technical drawing consists of two views of a mechanical part. The left view is a cross-section showing a stepped shaft with a central hole. Dimensions include: md1 (total length), md2 (length of the bottom step), md3 (length of the middle section), md4 (length of the top step), md5 (total length), td1 (bottom step thickness), td2 (middle section thickness), and td3 (top step thickness). The right view is a top view of a circular flange with a central hole and four smaller holes labeled 1, 2, 3, and 4. A red dashed line indicates the center of the flange.

MINESUP		HND Mechanical Fabrication		Part:		
Session: 2022		PHASE CONTRACT		Material: C30		
Phase NO: 30		Machine tool used: CONTROL POST				
Referential:						
MACHINING OPERATIONS		TOOLS		CUTTING ELEMENTS		
Ref.	Description	Cutting	Verification	Cs m/mm	N Rev/min	F mm/rev
a)	MACHINED F1F,F2F,F3F	MORTISE TOOL	Vernier calliper 1/50	27	47.77	0.1
	td2=5 , md1=37					

MINESUP		HND Mechanical Fabrication		Part: Material: C30		
		PHASE CONTRACT				
Session: 2022						
Phase NO: 50		Machine tool used: CONTROL POST				
Referential:						
MACHINING OPERATIONS		TOOLS		CUTTING ELEMENTS		
Ref.	Description	Cutting	Verification	Cs m/mm	N Rev/min	F mm/rev
a)	FINAL CONTROL		Vernier calliper 1/50			

CHAPTER FOUR

SUMMARY, CONCLUSION AND RECOMMENDATION

4.1 SUMMARY

The internship period at NOUGNING SARL was a period full of experiences hence we are not yet at top level, but we guess more and more experience will be acquire with time. It was involved with so many studies and activities which are cited in this document. This report is divided into several parts that involves the introduction, the various chapters and the conclusion.

The introduction mainly defines what an internship is and how the work is been organized . Chapter 2 in the logic shows the activities the intern did either alone or being assisted or participated. It also the problem encountered by the intern the benefits from the internship period. The chapter 3 concerns with the analysis of the case study of the intern. It contains all the tables of machining processes, phase contracts etc..... Hence analysis of the flange is well elaborated in this document. The last chapter which is chapter 4 shows the summary and the recommendations.

4.2 RECOMMENDATIONS

Base on the analysis and conclusions in this report, it's recommended that NOUGNING SARL should however improve on some few aspects in other to achieve their goal such as:

- They have to get new machines for more efficient jobs.
- The management should respect the preventive maintenance procedures in other to prevent break down of machines.
- Correct follow up safety procedures to prevent accidents and near miss in the workshop.
- Allocate their different interns in their specific field of study .

Conclusion

The internship period is very important to any student of the higher education since this helps the intern to blend theory to practical. This document shows how important it is to the intern as chapter2 shows the weekly activities within that period. Although several activities were done, the intern can't obtain total maturity and experience in this field and any other. Even though we encountered few problems during this training especially in the manufacturing of the flange, we were encourage that we completed the task and also learn a lot of new techniques and gain more experiences. Thus, we gladly finished the internship in this company, we hope this will pave our road in becoming great engineers in the future.

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APPENDICES

Appendix 2: Table of roughness value

TABLE OF ROUGHNESS VALUE

Surface roughness values produced by common production process and materials



Note: the ranges shown above are typical of the processes listed.
 Higher or lower values may be obtained under special conditions.

Figure 7: TABLE OF ROUGHNESS VALUE

Appendix 3: Symbolisation for normal location

SYMBOLS FOR NORMALS LOCATION (SECOND PART OF THE NORMS)			
d-Technological type of element		c- Nature of contact whit piece	
b- Nature of the surface		a- Function of the technological element	
a	Functions of technological element	Symbols	projected Representation
Accurate positioning Dimensioning start point Definition of an axis		BLACK TRIANGLE 	Centring  support  Complet  released
Blocking of the piece Pre-localisation Opposition to deformation and vibration		white Triangle 	
b	Nature of surface	Symbol	
MACHINED surface			(single line) 
ROUGH surface			(two lines) 
c	Symbols indicating the nature of contact with the surface		
Flat Touch		Serrated Touch 	curved Touch 
Articulation control		Released Touch 	Dish  Vee 
d	Symbol of technological types of elements		
Fixed support			Concentric clamping 
Fixed centring			Non-return support 
Claping system			Reversible support 

Figure 8:SYMBOLISATION OF NORMAL LOCATION

CUTTING SPEEDS (TURNING)

Material being machine	HSS tools		Carbide tools
	roughing	finishing	
Steels $R_m \leq 70$	22	26	90
Steels $70 < R_m \leq 100$	18	22	70
Steels $100 < R_m \leq 120$	16	20	60
EN-GJL 200	22	26	70
EN-GJS	16	20	60
CuZn	60	80	220

Appendix 4: Table defining the number of minimum operations expected

TABLE DEFINING THE NUMBER OF MINIMUM OPERATIONS EXPECTED					
NF E-04 030312					
TYPE OF FUNCTIONAL REQUIREMENT	PRECISION LEVEL	NUMBER OF OPERATIONS			
		1	2	3	4
		F	R+F	R+F/2+F	R+F/2+F+SF
INTERVAL OF TOLERANCE (LT)	$LT > 0,4$	x			
	$0,15 < LT \leq 0,4$	*	x		
	$0,05 < LT \leq 0,15$		*	x	
	$LT \leq 0,05$			x	x
IT/QUALITY (Q)	$Q \geq 12$	x			
	$Q = 9 - 10 - 11$	*	x		
	$Q = 7 - 8$		*	x	
	$Q \leq 6$			*	x
ROUGHNESS (Ra)	$Ra > 6,3$	x			
	$3,2 \leq Ra \leq 6,3$		x		
	$Ra < 3,2$			x	x
CODING OF ELEMENTARY SURFACES					
R : Roughing F/2 : 1/2 Finishing F : Finishing SF : Super finishing					
NUMBER OF ELEMENTARY SURFACES					
1 : F 2 : R + F 3 : R + F/2 + F 4 : R + F/2 + F + SF					
	: Unnecessary application	*	: less frequent application	x	: Average application

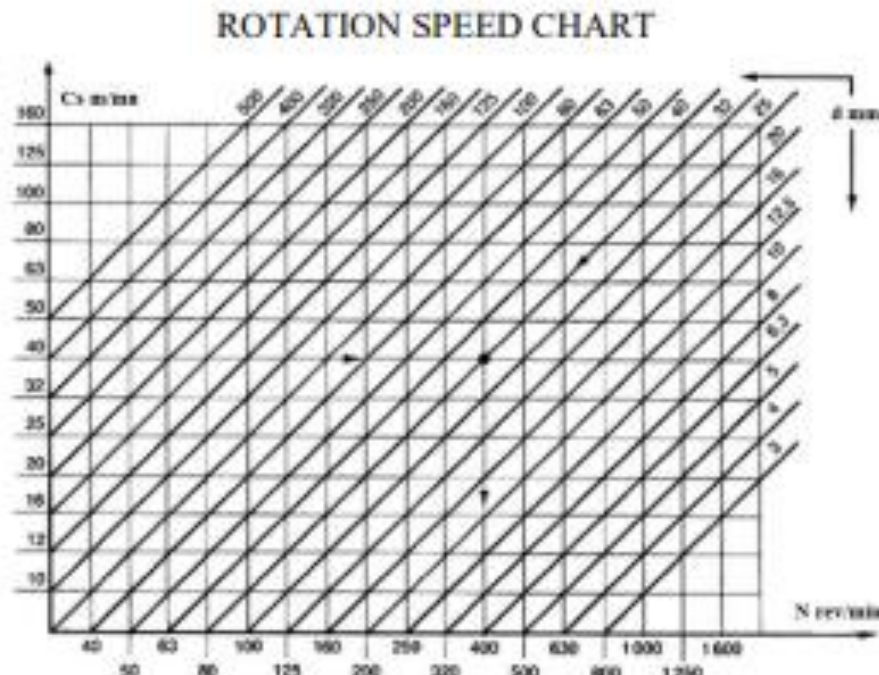


Figure 9: APPENDIX 4

Appendix 5: 3D drawing of the flange

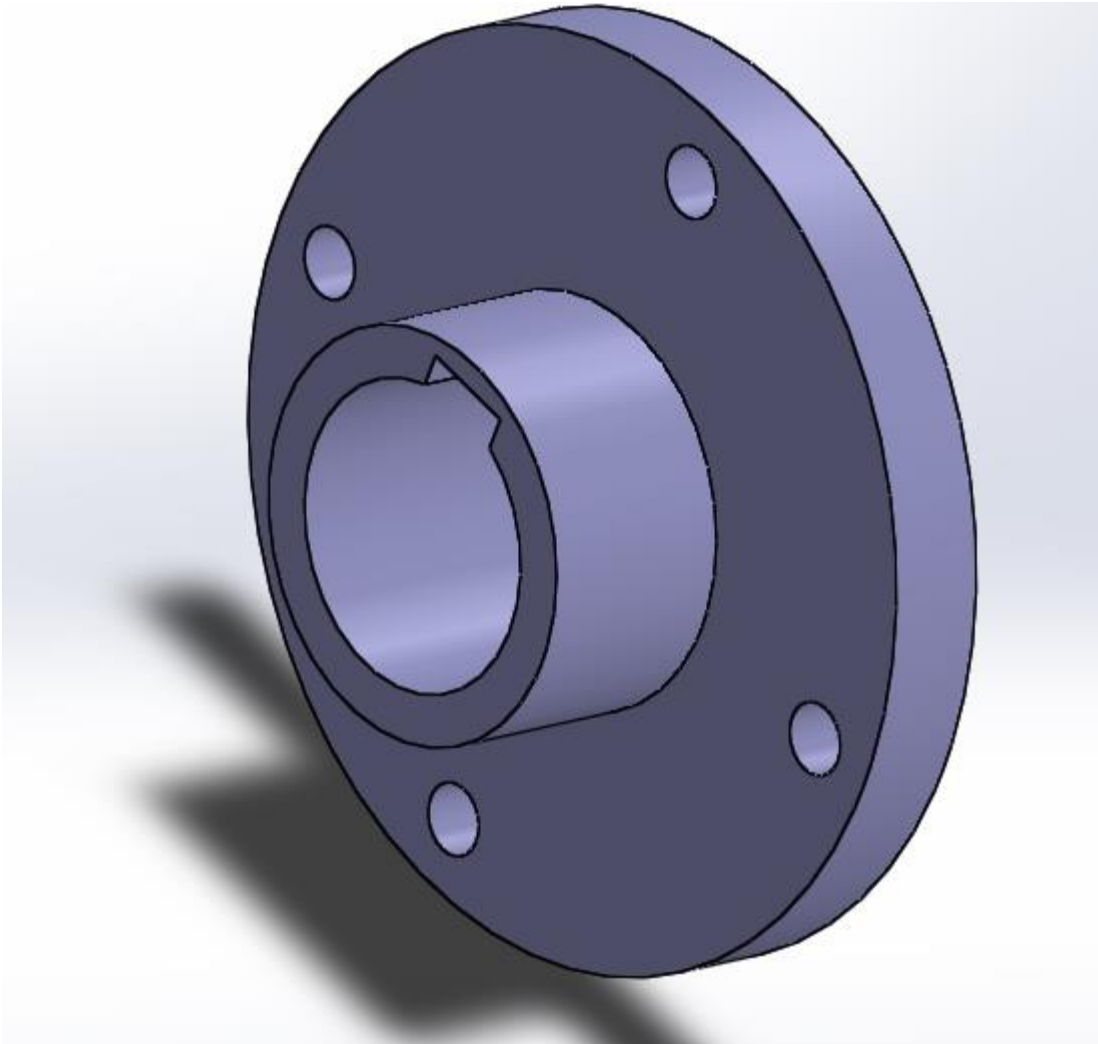


Figure 10:3D DRAWING

APENDIX 6: View of the concrete mixing machining



Figure 11: views of concrete mixing

APENDIX 7: lathe machine

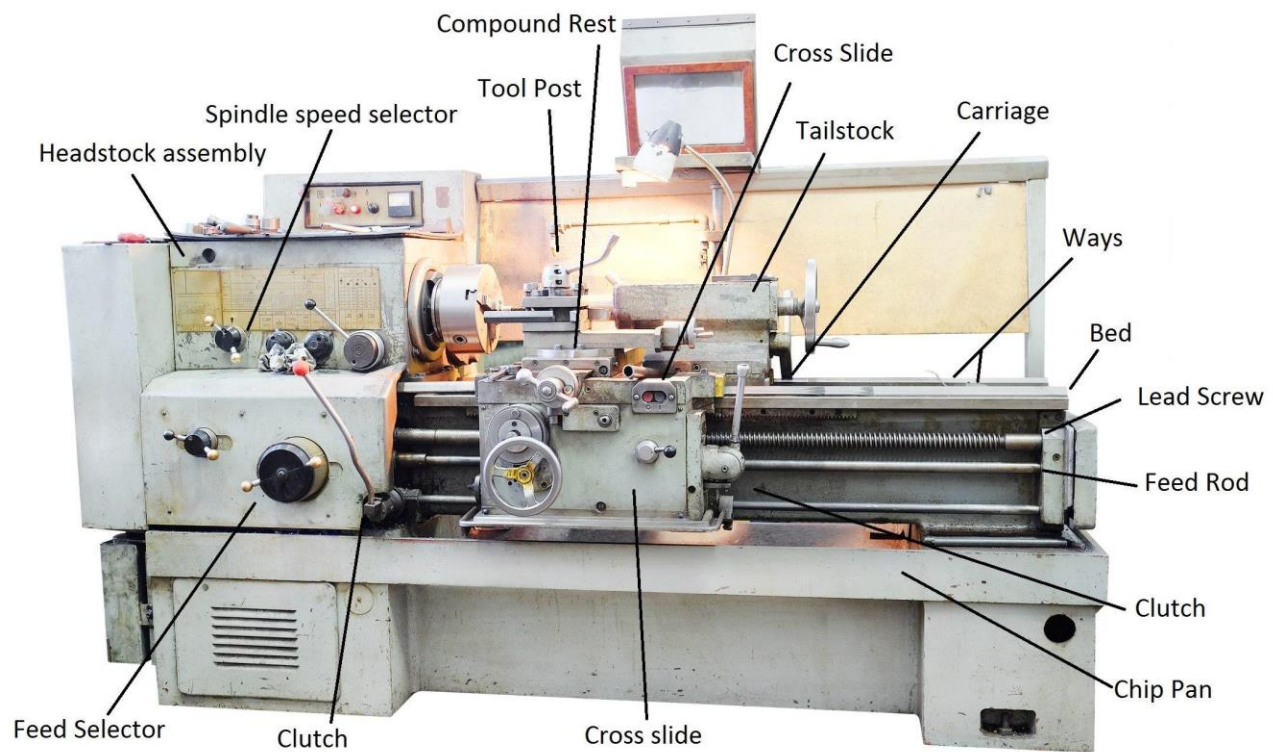


Figure 12: LATHE MACHINE

APENDIX 8 : Vernier caliper

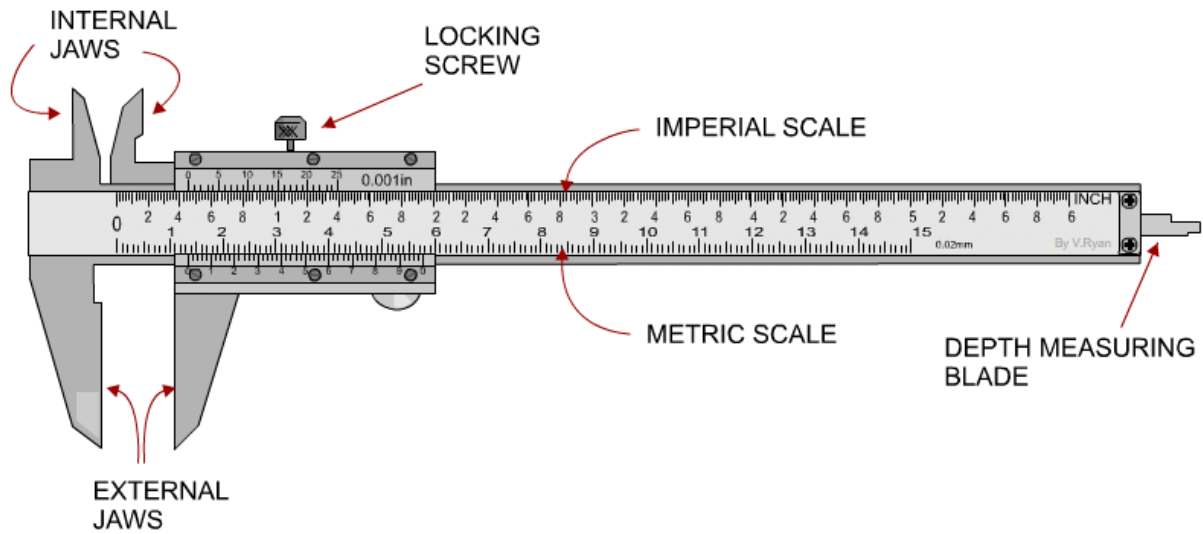


Figure 13: VERNIER CALIPER